



The health consequences of aerial spraying illicit crops: The case of Colombia[☆]



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ABSTRACT

This paper exploits variations in aerial spraying across time and space in Colombia and employs a panel of individual health records in order to study the causal effects of the aerial spraying of herbicides (glyphosate) on short-term health-related outcomes. Our results show that exposure to the herbicide used in aerial spraying campaigns increases the number of medical consultations related to dermatological and respiratory illnesses, as well as the number of miscarriages. These findings are robust to the inclusion of individual fixed effects, which compare the prevalence of these medical conditions for the same person under different levels of exposure to the herbicide used in the aerial spraying program over a period of 5 years. Also, our results are robust to controlling for the extent of illicit coca cultivation in the municipality of residence.

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1. Introduction

One of the main strategies used in Colombia to fight illegal drug production and reduce the supply of cocaine is that of the aerial spraying of herbicides on coca crops, the raw material used to produce cocaine.³ Under Plan Colombia,⁴ the average number of

hectares sprayed annually with herbicides during the last decade is 128,000.⁵ At its peak in 2006, 172,000 ha were aerially sprayed with glyphosate, the herbicide used in the aerial spraying program in Colombia. The effectiveness of this approach has been thoroughly defended by the US, yet attacked and questioned by NGOs and opponents of the so-called “war on drugs”. Nevertheless, the debate about the effectiveness of aerial spraying campaigns and their collateral effects is often grounded in ideologies and rarely takes the available scientific evidence seriously. However, both structural evaluations (Mejía and Restrepo, 2016) and reduced-form estimation techniques (Reyes, 2014; Roza, 2014 and Mejía et al., 2015) have exploited exogenous sources of variation to assess the impact of aerial spraying campaigns on the reduction of coca cultivation in Colombia, consistently indicating that the effects are very small. On top of its limited effectiveness, this “chemical war” (as it is often called by opponents of the war on drugs) has been linked with all sorts of collateral negative effects. Examples include distrust felt towards state and government institutions by affected populations,⁶ non-negligible negative effects on the environment (especially on amphibian populations through the contamination

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³ Poppy seeds in the case of heroin production and coca bushes in the case of cocaine production.

⁴ Plan Colombia is the name of a joint strategy launched in 2000 between the governments of Colombia and the US for the fight against illegal drugs and organized crime.

⁵ 128,000 ha correspond to 0.3% of the total arable land in Colombia. These calculations were done with information from the 2014 National Agricultural Census.

⁶ See Felbab-Brown (2009) and García (2014), Landy (1988), Navarrete-Frías and Thoumi (2005), among others.

of water sources),⁷ and negative health effects for affected populations exposed to the herbicides. However, it is important to note that most of the evidence regarding these collateral effects comes from fieldwork, with few observations having been made, thus generating problems of internal and external validity. Both fieldwork and anecdotal evidence on the effects of aerial spraying on health are plagued by confounding factors that make it difficult to link aerial spraying as a direct cause of the aforementioned maladies. One of the most prominent confounding factors is that of coca cultivation itself. More precisely, given the high spatial correlation between coca cultivation and the occurrence of aerial spraying campaigns, it can be easily argued that if coca cultivation and cocaine production themselves make indiscriminate use of pesticides and other agricultural inputs and chemical precursors, then these activities may be the ones generating the negative health and environmental consequences observed by researchers in the field. In short, most of the evidence (anecdotal, from field work and empirical) is plagued by issues of endogeneity and omitted variables that make it hard to reach causal conclusions regarding the effects of spraying campaigns on health outcomes.

In this paper, we use a large administrative panel data set collected by the Ministry of Health that contains individual health records across a five-year period. This data allows us to observe individuals more than once during the period of analysis. Therefore, our identification strategy relies on the fact that we can construct a panel of individuals and include fixed effects at this level. As thus, we are able to control for individual unobservable characteristics that do not change over time, such as baseline health. We merge these health records together with precise information on the location and exact timing of aerial spraying events (at a daily and municipal level) in order to disentangle the causal effects of the aerial spraying of herbicides on a broad range of health outcomes. Our identification strategy tackles the problems of endogeneity present in the previous literature since it relies on the fact that each aerial spraying event is neither announced nor anticipated. Running a regression model we are able to reject the hypothesis that there is a predictable pattern for aerial spraying over time.⁸ Therefore, we assume that individuals cannot anticipate with certainty the time and extent of exposure to aerial spraying campaigns. This enables us to exploit a quasi-natural experiment to test the causal impact of exposure to glyphosate on human health. Moreover, in our view this quasi-experiment is the closest we can get to a randomized experiment (which, for obvious reasons, would be impossible to implement in the context of human exposure to glyphosate).

On the one hand, our health data contains the individual-level registries of medical consultations for more than 76 million (individual-time) observations. This is an unbalanced panel that covers a period of five years. On the other hand, we have official records from the Colombian National Antinarcotic Police for the number of square kilometers sprayed daily per municipality over a period of five years, that is, between January 2003 and December 2007. These are precisely the years that evidence the highest levels of aerial spraying during the last 15 years under Plan Colombia.

Combining these two panel data sets we estimate individual fixed-effects regressions that test whether there exists an increase in the probability of experiencing a health problem related to herbicide exposure. The fixed effect model captures the exposure to different levels of aerial spraying for the same individual in different moments in time. We also include month and year fixed effects that control for seasonal illnesses or harvesting seasons. Our findings coincide with the medical literature and robustly indicate that the aerial spraying of glyphosate increases the probability of having dermatological and respiratory problems, as well as miscarriages (Sanborn et al., 2012; Sanborn et al., 2007; Cox 1995a; Sherret, 2005; Regidor et al., 2004; Solomon et al., 2007). These results are robust to different specifications of the empirical model and to the inclusion of a wide range of controls, including the extent of coca cultivation at the municipality level. It is important to highlight that given the nature of our data we are unable to capture health consequences in the long-term that might translate into reduced life expectancy, quality of life or productivity.

We run a placebo test in which we take as the dependent variable the proportion of two conditions that, in principle, should be completely unrelated to exposure to the spraying campaigns: accidents and bone fractures.⁹ As expected, the results show that exposure to the spraying campaigns does not lead to an increase in the fraction of medical consultations related to these two diagnoses. In addition, we perform a placebo test for timing of exposure by opening the window onto a longer period of time. The results show that the effect loses significance or disappears when using an inappropriate time window.

Our paper offers four main strengths and contributions in relation to the existing literature. First, to the best of our knowledge, this is the first paper in the literature that uses a quasi-experiment to estimate the effects of the spraying of illicit crops on various health outcomes in a drug-producing country. In particular, given that the exact timing and magnitude of spraying campaigns are unannounced, it is difficult to anticipate spraying events are arguably an exogenous shock from an individual's point of view, and this strengthens our identification strategy and the internal validity of our results. Second, the large sample size of our dataset also allows us to find robust and precise results in the econometric specifications, even if the actual effects are small. We use a dataset that contains administrative records for all of the health service institutions in Colombia from 2003 to 2007, accounting for more than 76 million visits to the doctor and approximately 3 million completed and non-completed birth registrations. By using information from across the entire Colombian population, our results exhibit greater external validity than those performed in the field or in laboratories by epidemiologists or medical doctors. Third, our daily data is appropriate for the establishing of a precise link between the date and magnitude of aerial spraying, and the date on which individuals go to the hospital to see the doctor or visit the emergency room. Finally, from the administrative health records used we are able to construct a panel for individuals that go to a health service provider more than once during our period of analysis. The possibility of comparing the same individual over time, by estimating an individual fixed-effects model, isolates all instances of genetic, behavioral, and other time-invariant, unobserved individual heterogeneities. This automatically rules out a range of confounding factors from our study and omits the variable biases present in cross-sectional studies.

⁷ See Cox (2004), Imming (2010), Navarrete-Frías and Veillete (2005) and Relyea et al. (2006) for studies documenting the effects of aerial spraying with glyphosate on the environment (deforestation, pollution of water sources, etc.) and animal species.

⁸ We run a regression for the daily amount of aerial spraying on different time variables, which include: day of the week, day of the month, month of the year, and year. This regression tests whether spraying has a predictable pattern over time. We find no statistical difference in the average sprayed in the day of the month, and no difference in spraying between days of the week, except for Wednesdays when a lower amount of spraying takes place on average. Regarding month of the year, May, June and July, together with November and December, exhibit a lower amount of spraying on average. With these results we are confident that, at least on a daily basis, aerial spraying is an unexpected or unanticipated shock.

⁹ We define accidents by using the following categories from the ICD10 diagnoses: traumatism and fractures (codes starting with S), injuries and collisions (V codes), falls, bites or stings (W codes), burns or poisoning (X codes). Bone fractures comprise some specific codes of the S category and so represent a subset of accidents.

The paper proceeds as follows. Section 2 reviews the relevant medical and epidemiological literature related to the effects of exposure to herbicides on health outcomes and outlines our paper's contribution to the relevant literature. Section 3 presents the Colombian context and describes the aerial spraying program. Section 4 describes in detail the data used in our empirical exercise. Section 5 discusses the basic theoretical framework behind our empirical strategy. Section 6 reports and analyzes the main results. The final section presents the concluding remarks.

2. Medical and epidemiological literature on the collateral effects of glyphosate on health outcomes

A wide variety of medical studies have documented the negative impacts of exposure to glyphosate on human health, yet a definitive consensus remains absent from the literature. Therefore, this paper aims to provide answers to an ongoing debate. Prior medical literature primarily consists of cross-sectional studies that compare the prevalence of health outcomes among those with and without previous exposure to herbicides (especially glyphosate) or laboratory animal experiments. Sanborn et al. (2012) and Sanborn et al. (2007) argue that herbicide use causes dermatological problems. They highlight multiple cross-sectional studies that demonstrate a higher prevalence of burns, irritations and skin redness in exposed groups. Experimental evidence for animals and observational studies with humans subjected to accidental skin contact with glyphosate reviewed by Cox (1995a) find redness, swelling and irritation of the skin after exposure. Sherret (2005) reports anecdotal evidence of respiratory ailments following aerial spraying campaigns in Colombia, including bronchial irritation. This is corroborated by some experimental evidence on glyphosate inhalation in animals, as reviewed by Cox (1995b). However, a review by Mink et al. (2011) argues that current medical literature fails to display a consistent pattern between glyphosate exposure and respiratory conditions.

Some researchers have addressed the effects of herbicide exposure on miscarriages and fetus malformation. Laboratory studies evidence the toxicity of glyphosate for human reproductive cells. For example, Benachour and Seralini (2009) find that exposure to glyphosate causes premature death in umbilical, embryonic and placental cells at low doses similar to the residues found in herbicide-treated food. Animal studies documented in Cox (1995b) show decreased female fertility and lower birth weights following the ingestion of products that have been sprayed with glyphosate. Sanborn et al. (2012), Regidor et al. (2004), Sanborn et al. (2007), and Solomon et al. (2007) indicate that exposure to herbicides before conception is correlated with miscarriages during the first trimester of pregnancy. They also find that direct exposure of the father to large amounts of herbicides is linked to a greater risk of fetal death, an effect that increases if exposure took place within a three-month period preceding conception. Cases of anencephaly associated with direct exposure of the mother to herbicides during the preconception period (between the three months preceding conception and the third month of pregnancy) are also reported in the medical literature. Arbuckle et al. (1999) finds an association between exposure to herbicides in the three months prior to conception and the month of gestation with spontaneous abortions at 20 weeks of gestation or less. This evidence is also supported by Bretveld et al. (2008), who find an increased risk of abortion among female greenhouse workers. Furthermore, Sanborn et al. (2012) provide evidence of an association between the spraying of herbicides and problems of fecundity and lower sperm concentration. In contrast, Williams et al. (2012) cite cross-sectional and laboratory studies that suggest that glyphosate exposure is not related to miscarriages or developmental effects at the exposure levels of her-

bicide use. Similarly, Solomon et al. (2007) indicate that the levels of concentration of glyphosate used in aerial spraying campaigns in Colombia are so low that they don't carry a significant risk to human health.

Sanborn et al. (2012) and Sanborn et al. (2007) find minor evidence for negative effects following exposure to glyphosate on levels of depression, anxiety, neural disorders, minor tact sensibility, abnormal reflexes and psychomotor dysfunction. As for neurodegenerative diseases, they find evidence of a connection between exposure to pesticides at work and Parkinson's and Alzheimer's later in life.

Finally, a recent review finds limited evidence in humans for the carcinogenicity of glyphosate (International Agency for Research on Cancer (IARC), 2015). More precisely, a review of the available evidence finds that exposure to the herbicide leads to an increased risk of non-Hodgkin lymphoma. These findings led the IARC to reclassify glyphosate as category 2A: probably carcinogenic to humans.

The contribution made by our paper to the literature relies on the strong emphasis that we place on performing a clean identification strategy. To the best of our knowledge, this is the first paper that exploits a quasi-natural experiment to address the potential endogeneity issues that arise when estimating the effects of the aerial spraying of illicit crops with herbicides on human health outcomes. While our results corroborate some of the results found in the medical literature (e.g., the negative effects of exposure to glyphosate on dermatological and respiratory problems, as well as on miscarriages), our dataset and empirical strategy do not allow us to study the health effects on other health conditions for which it is difficult to precisely pin down the time that elapses between exposure to the herbicide and the appearance of any symptoms. In addition, this paper complements the results on miscarriages included in the Johns Hopkins – Lancet commission on Drug Policy and Health (Csete et al., 2016) by providing a detailed explanation of the construction of the data and the empirical strategy.

The definition of the window of exposure prior to consultation is defined according to the findings outlined in the guides for the adequate use of pesticides from the University of Kentucky, College of Agriculture and from the University of Missouri (Fisher and Andre, 2002; University of Kentucky-College of Agriculture, 2017). Regarding dermatological diagnoses, there exists evidence that effects appear 24 h after exposure to glyphosate. With regard to respiratory diagnoses, there is evidence that effects appear between 3 and 14 days after exposure. In our estimations we use greater time windows to account for the time that individuals take to recognize their health condition and find some time to plan to go to the doctor. Therefore, we consider spraying done 7 and 15 days prior to consultation for dermatological and respiratory diagnoses, respectively.¹⁰

3. Colombian context

3.1. The aerial spraying program¹¹

Following the large increase in coca cultivation that took place in Colombia during the second half of the 1990s and the increasing involvement of the Fuerzas Armadas Revolucionarias de Colombia (FARC) and paramilitary groups in this illegal business, in September 1999 the governments of Colombia and the US launched a joint strategy which would come to be known as Plan Colombia. According to an official report by the US Government Accountability Office (GAO), US funding for the military compo-

¹⁰ We also undertook various robustness checks with different time windows, as explained in detail in the empirical section.

¹¹ This section is adapted from Mejía et al. (2015).

Table 1
Descriptive statistics for the dermatological and respiratory panels in different subsamples.

Variable	Obs	Mean	Std. Dev.
Panel A: Complete sample			
Proportion dermatologic	39,766,259	0.012	0.082
Proportion respiratory	39,766,259	0.036	0.135
Daily average spraying in the last 15 days	39,766,259	0.003	0.077
Daily average spraying in the last 7 days	39,766,259	0.004	0.096
Proportion of coca cultivation in municip.	39,766,259	0.0002	0.003
Panel B: Sample of municipalities with positive levels of aerial spraying			
Proportion dermatologic	7,264,691	0.014	0.086
Proportion respiratory	7,264,691	0.039	0.139
Daily average spraying in the last 15 days	7,264,691	0.019	0.180
Daily average spraying in the last 7 days	7,264,665	0.020	0.225
Proportion of coca cultivation in municip.	7,264,691	0.001	0.007
Panel C: Sample of municipalities with non migrant population ^a			
Proportion dermatologic	37,736,485	0.012	0.082
Proportion respiratory	37,736,485	0.036	0.135
Daily average spraying in the last 15 days	37,736,485	0.003	0.078
Daily average spraying in the last 7 days	37,736,485	0.004	0.097
Proportion of coca cultivation in municip.	37,736,485	0.0002	0.003
Panel D: Sample of High income municipalities ^b			
Proportion dermatologic	20,131,375	0.012	0.0817
Proportion respiratory	20,131,375	0.034	0.1304
Daily average spraying in the last 15 days	20,131,375	0.0003	0.0086
Daily average spraying in the last 7 days	20,131,375	0.0003	0.0128
Proportion of coca cultivation in municip.	20,131,375	0.0000	0.0001
Panel E: Sample of Low income municipalities ^c			
Proportion dermatologic	19,634,884	0.012	0.082
Proportion respiratory	19,634,884	0.039	0.140
Daily average spraying in the last 15 days	19,634,884	0.007	0.109
Daily average spraying in the last 7 days	19,634,884	0.007	0.136
Proportion of coca cultivation in municip.	19,634,884	0.0005	0.004

^a We define non-migrant population as individuals that appear in the same municipality for the whole period of study.

^b We define high-income municipalities as those above the 50th percentile of tax income collected by the municipality.

^c We define low-income municipalities as those below the 50th percentile of tax income collected by the municipality.

nent of Plan Colombia was, on average, US\$540 million per year between 2000 and 2008. The Colombian government, for its part, invested approximately US\$812 million per year to fight drugs and drug-related organized crime. Summing up these expenditures represents approximately 1.2 percent of Colombia's average annual GDP between 2000 and 2008. As such, Plan Colombia is the largest anti-drug intervention that has ever been implemented in a drug-producing country.

Plan Colombia developed two types of antidrug policy: those directed towards the reduction of production and those directed towards fighting trafficking. The first group of policies sought to undermine coca leaf production via the aerial spraying of herbicides and the manual eradication of crops. Aerial spraying has been by far the main anti-drug strategy in terms of financial resources invested. On average, 128,000 ha have been sprayed with herbicides per year, reaching a peak of 172,000 ha sprayed in 2006. Manual eradication procedures have accompanied aerial spraying since 2005. Other complementary strategies aimed at reducing the cocaine supply include controlling the chemical precursors used in the processing of the coca leaf into cocaine, as well as the detection and destruction of cocaine-processing laboratories. The second group of policies, which were used to fight trafficking, strived to detect and block the distribution of drugs (Gaviria and Mejía, 2016).

American contractors, such as DynCorp, carry out spraying campaigns in small aircraft. Coca crops are sprayed with substances such as Roundup, which contains glyphosate as its main active ingredient. The herbicide also contains the surfactant POEA, which helps the glyphosate to penetrate the coca plant's foliage. This herbicide inhibits the enzyme involved in the synthesis of the aromatic amino acids, thus killing the plant. Glyphosate is absorbed through the foliage and is only effective on actively growing plants (that

is to say, it is not effective in preventing seeds from germinating). Though Roundup, the commercial name of the herbicide, was designed to kill weeds and grasses, including coca bushes, it may also affect other legal crops that are non-resistant to glyphosate. However, aerial spraying with glyphosate is targeted at areas where coca crops have been detected using satellite images, implying that areas with coca crops are much more likely to be sprayed and destroyed by this enforcement strategy. As a result, farmers who decide to grow coca bushes face the risk of their crops being destroyed by the herbicide used in aerial spraying campaigns. In the face of aerial spraying, farmers may still grow coca bushes and play their luck, or mitigate the effects of the herbicide using a variety of techniques. For instance, farmers who know that an area is being targeted by the program sometimes spray molasses on the coca bushes to prevent the herbicide from penetrating the foliage and killing the plant. Also, if farmers cut the stem of the plant a few hours after the spraying event, the plant grows back again and can be harvested a few months later. However, these alternatives are costly, and other farmers may decide to relocate their crops to areas less likely to be sprayed or to start cultivating solely legal crops that are not targeted by the spraying campaigns. Economic theory suggests that in the face of higher risk from the spraying of herbicides, marginal farmers should either relocate their coca crops or reduce their cultivation, while other farmers would continue to grow coca nonetheless.

3.2. The health system in Colombia

Health insurance coverage among the Colombian population is above 90 percent. There are two types of health insurance regime: subsidized and contributory. The first regime covers the poor and

Table 2
Descriptive statistics for the miscarriages panels in different subsamples.

Variable	Obs	Mean	Std. Dev.
Panel A: Complete sample			
Miscarriages	2,947,070	0.164	0.370
Daily average aerial spraying in last 9 months	2,947,070	0.004	0.040
Proportion of Coca cultivation in municipality	2,947,070	0.0004	0.004
Panel B: Sample of municipalities with positive levels of aerial spraying for non-migrants			
Miscarriages	585,132	0.168	0.374
Daily average aerial spraying in last 9 months	585,132	0.020	0.089
Proportion of Coca cultivation in municipality	585,132	0.002	0.009
Panel C: Sample of municipalities with non migrant population ^a			
Miscarriages	2,815,385	0.163	0.370
Daily average aerial spraying in last 9 months	2,815,385	0.004	0.041
Proportion of Coca cultivation in municipality	2,815,385	0.0004	0.004
Panel D: Sample of High income municipalities ^b			
Miscarriages	1,415,192	0.158	0.3651
Daily average aerial spraying in last 9 months	1,415,192	0.0002	0.0018
Proportion of Coca cultivation in municipality	1,415,192	0.0000	0.0001
Panel E: Sample of Low income municipalities ^c			
Miscarriages	1,486,215	0.159	0.366
Daily average aerial spraying in last 9 months	1,486,215	0.0002	0.002
Proportion of Coca cultivation in municipality	1,486,215	0.0000	0.000

^a We define non-migrant population as individuals that appear in the same municipality for the whole period of study.

^b We define high-income municipalities as those above the 50th percentile of tax income collected by the municipality.

^c We define low-income municipalities as those below the 50th percentile of tax income collected by the municipality.

the unemployed, and is conditional on ranking low on the poverty index score or having a condition of vulnerability. The second regime is mandatory for all formal employees; both employer and employee pay a portion of this service as part of their payroll taxes. Reports from the Ministry of Health show that, by 2008, 88.85% of the Colombian population was covered by health insurance. Of those insured, 44% participated in the contributive regime and 56% in the subsidized regime. This amounts to coverage for 39 million people out of 43.9 million inhabitants.¹²

The copayments made for medical consultations are determined by a person's base income in the contributive regime. Individuals who earn less than two minimum wages pay less than 1 US dollar, those earning between 2 and 5 minimum wages pay 3.5 US dollars, and those with an income above 5 minimum wages pay 9.3 US dollars for every visit to the doctor. Individuals with subsidized health insurance do not have to make payments for medical consultations, but they do have make copayments of, at most, 10% of the value of hospitalization services with a defined cap. There is no difference in terms of length of medical consultation for the amount of copayment.

If individuals go to a medical consultation they have to see a generalist first, before being remitted to a specialist if necessary. If they go to an emergency room then they can see a specialist directly. Regardless of whether the patient is visiting a generalist or a specialist, the type of consultation is registered according to its main diagnosis and is not given by the type of specialist visited.

Even though a high proportion of the population has some kind of health coverage, it is important to acknowledge that there exists an amount of substitution of medical services with non-traditional medicine or self-medication. A doctor's prescription is necessary for the purchase of medicine in specific and few cases, for instance, antibiotics. In general, if people wish to buy medicines they are not asked for a prescription. According to the Demographic and Health Survey (DHS) 2005, of the population that needed health treatments only 1% of individuals went to informal medical ser-

vices, 29% went to a drugstore or self-medicated, and the remaining 70% attended formal medical services.

4. Data

4.1. Administrative registry of medical consultations, emergency room visits and hospitalizations

The Individual Register of Health Services Provision (RIPS, as per its acronym in Spanish) contains the individual-level registers for medical consultations, emergency room (ER) visits, hospitalizations and procedures that took place in any health service institution in Colombia between 2003 and 2007. Each registry contains information about the appointment (date, municipality of the facility visited, diagnosis according to the ICD-10,¹³ institution that treated the patient and consultation fee) and the patient (age, gender and type of health insurance). It is important to note that we do not have information for, nor can we infer anything about, individuals who did not attend a health service institution during our period of study. As we infer from the DHS 2005, this could account for approximately 30% of the population.

The quality of our administrative data is important for the credibility of our results. It is important to take into account several considerations regarding the cleaning process of our data. Table A1 in Supplementary material reports in column 2 the final number of observations in the RIPS database after the cleaning process for the years 2003–2007, which accounts for 13–16 million observations per year and 76 million records in total. Given that in our preferred specification we include individual fixed effects, we exclude observations that correspond to individuals that only appear once in the panel. We also exclude individuals with inconsistencies in age across the five-year period of study. We merge our resulting RIPS database with the municipal panel of aerial spraying and rainfall databases. This final database consists of 39,925,671 observations.

¹³ ICD-10 is the International Statistical Classification of Diseases and Related Health Problems (ICD), a medical classification list compiled by the World Health Organization (WHO). It contains codes for diseases, signs and symptoms, abnormal findings, complaints, social circumstances, and external causes of injury or disease.

¹² <https://www.minsalud.gov.co/salud/Paginas/CoberturasdelRégimenSubsidiado.aspx>

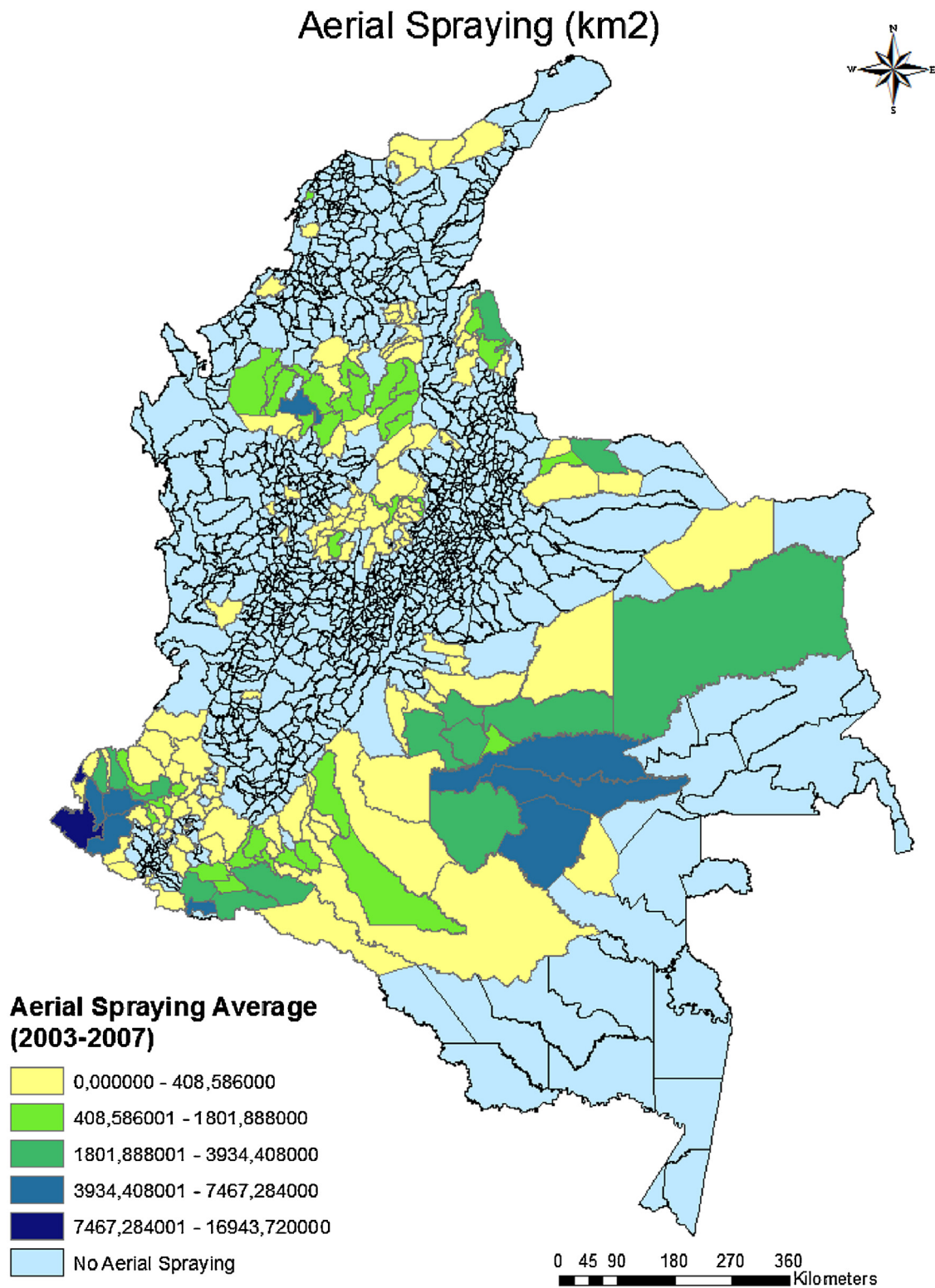


Fig. 1. Average square kilometers of Aerial Spraying in Colombia (2003–2007).

Source: Authors calculations using information from the Colombian Anti-Narcotics Police.

For further details regarding the cleaning process of the dataset, see Appendix A in Supplementary material. In addition to checking the consistency within the data, we use the DHS as an independent source of information to assess the number of people who visit health services in Colombia. In 2005, 13.2% of the population faced a health condition; among those people, 67.5% sought pro-

fessional assistance. In 2005, this represented roughly 4.3 million people who went to the doctor. This does not take into account the fact that an individual might go to the doctor more than once in a given year.

With the diagnoses reported in our medical consultation panel, and based on the medical literature summarized above, we con-

struct three groups of diagnostics potentially affected by the aerial spraying program: dermatological, respiratory and miscarriages. These three broad categories form the main focus of our analysis since we concentrate on health outcomes that appear in the short run (that is, a few days or months after exposure to the herbicide used in the aerial spraying program in Colombia). Other health-related problems that may take more time to develop are left out of our analysis since it is difficult to precisely identify the time between herbicide exposure and the appearance of certain medical conditions. Each group collects a variety of diagnoses selected using the ICD-10 codes and in accordance with the findings in the medical literature. Appendix B in Supplementary material includes a table that lists the diagnoses considered in the estimations, as well as the proportion that each represented in the total number of events in our panel.

Table 1 shows descriptive statistics for dermatological and respiratory variables used in our empirical exercise. Similarly, Table 2 shows descriptive statistics for miscarriages. These tables include the following five panels for different subsamples that will identify slightly different effects: complete sample (Panel A) as our baseline; sample of municipalities with positive levels of aerial spraying (Panel B) to capture the effect on the intensive margin; non-migrant population sample (Panel C) to capture any bias related to selective migration; sample of high-income municipalities (Panel D); and sample of low-income municipalities (Panel E) to identify if there are differences in the quality of the data and to test the validity of our results in two different samples. (i.e., under the assumption that high income municipalities have better health reports, we would only find the effect of aerial spraying for this sample). We create a dummy variable that equals 1 if the person appears in our panel with a diagnosis related to a dermatological or respiratory condition, and 0 if the person appears in the panel with some other diagnosis. We then use the number of visits to a health service institution in a year to construct the number of dermatological and respiratory diagnoses as a proportion of the total number of visits in a given year. As shown in Table 1, dermatological and respiratory consultations related to aerial spraying correspond to approximately 1.2% and 3.6% of all reported diagnoses, respectively.

With respect to miscarriages, we expect them to be severely underreported in our panel since many do not take place in a health service institution. Therefore, using the RIPS dataset we construct a proxy variable for miscarriages using information from prenatal care visits, hospitalizations and emergency room visits related to births. We assume that a miscarriage occurred if we see that a mother attended a prenatal care visit in a health service institution, but we do not find a birth registered in the hospitalization sample in the following 10 months.¹⁴ However, we should stress that, on the one hand, our proposed measure may overstate the number of miscarriages, since it does not take into account the fact that some women might attend prenatal care visits and then give birth in locations different from those of hospitals, clinics and other formal medical institutions. Also, our measure assumes that prenatal care visits and all births are reported. On the other hand, our measure may underestimate miscarriages given that it does not take into account miscarriages that take place during the first months of pregnancy, when the mother perhaps did not know about the pregnancy and did not attend a prenatal care visit. Although this variable will not give us a precise measure for the number of miscarriages,¹⁵ the correlation between our constructed measure of miscarriages

and reported miscarriages from the vital statistics at the municipal level is above 90%, making it a good proxy variable. An important control to include in the prediction of miscarriages is whether it is a first pregnancy; women who have had a previous pregnancy may be less likely to attend a prenatal care visit. Unfortunately, we do not have this information, but we are able to determine the first pregnancy within our period of analysis and control for it. The results of the regression that include this control are similar to those presented in the paper.¹⁶

4.2. Aerial spraying and coca cultivation data

We have very detailed information for each aerial spraying of illicit crops event for our period of study. The data for spraying campaigns is recorded by geo-coding devices that are built into the aircrafts used for the aerial spraying campaigns. Information on the location (municipality) and number of hectares sprayed is recorded during the flight and then collected by the national authorities when the plane lands. This information was obtained from the Colombian Anti-Narcotics Police, a special unit inside the Colombian National Police in charge of designing and implementing most of the strategies used in the fight against illegal drug production and trafficking in the country. These records include the exact date and time of the spraying event, the municipality of occurrence, the number of hectares sprayed and the type of illicit crop sprayed (coca, marijuana or opium poppy). It is worth mentioning that we only use the spraying of coca crops, and not those of marijuana or the opium poppy. However, this is not a major concern since the spraying of these crops is very small compared to that of coca crops. In 2002, coca crops accounted for 98.3% of spraying and in 2007 it accounted for all of the spraying undertaken.

Information about coca cultivation was obtained from SIMCI, the United Nations Office in Colombia in charge of measuring the extent of coca cultivation. Coca cultivation is measured using satellite images that are taken at the end of each calendar year. The satellite pictures cover the entire Colombian territory and are analyzed by computer programs and experts to precisely identify and distinguish coca crops from other crops (legal or illegal). Fig. 1 presents a map of the average square km aerial spraying of coca crops in Colombia during our period of study, 2003–2007.

It is very important to clarify that all of our estimations control for the extent of coca cultivation at the municipality level. If coca cultivation and aerial spraying are positively correlated (as is indeed the case), then controlling for the level of coca cultivation is important to prevent the estimation of our coefficient of interest from being biased. If we did not control for coca cultivation, then the health effects of exposure to chemical precursors used in the process of transforming the coca leaf into the coca base would be captured by our coefficient of interest (namely, that which accompanies aerial spraying).

Tables 2 and 3 report descriptive statistics for the aerial spraying daily mean according to the relevant exposure time window (in square kilometers), and the proportion of municipal area cultivated with coca for the four subsamples. We define the relevant time windows of glyphosate exposure according to the findings in the literature for each specific health condition. Specifically, we use seven days for dermatological conditions,¹⁷ fifteen days for respi-

Therefore, our estimates would not be biased; standard errors would be higher and statistical significance lower.

¹⁶ These regressions are available upon request.

¹⁷ The literature finds that the effects of dermatological problems can appear 24 h after exposure. We therefore use a longer time window of 7 days for dermatological events to account for the time that a person takes to acknowledge that he/she has a medical issue and plan to go to the doctor. Furthermore, the results are positive and significant with any time window between 3 and 15 days.

¹⁴ A child born at full term (40 weeks) has spent 9.3 months in the womb. Some babies are born at 42 weeks, and so we calculated that 10 months accounts for the full period of pregnancy in these cases.

¹⁵ This issue might generate classical measurement error in the dependent variable. If this were to be the case it would be uncorrelated with aerial spraying.

Table 3
Restricted to common support on area of the municipalities.

Variables	Non-sprayed		Sprayed		Differences	p-value
	N	Mean	N	Mean		
Proportion Dermatologic	632	0.012	106	0.013	0.0002	0.626
Proportion Respiratory	632	0.041	106	0.039	−0.002	0.256
Proportion Miscarriages	626	0.205	109	0.202	−0.003	0.704
Proportion Fractures	632	0.011	106	0.010	−0.001	0.083
Proportion Accidents	632	0.012	106	0.010	−0.002	0.073
Age	632	30.628	106	27.041	−3.587	0.000
Male (%)	632	0.366	106	0.366	−0.0004	0.924
Population	632	47494.26	106	61129.84	13635.58	0.654
Area (km squared)	632	487.043	106	1254.283	767.240	0.000
Rural index	620	2.441	106	3.095	0.654	0.085
SGP-health per capita	632	69.154	106	66.005	−3.149	0.238
Tax Income per capita	632	0.013	106	0.014	0.001	0.651

ratory problems, and a window of nine months (corresponding to eight months prior to the last prenatal care visit and one month later) for miscarriages.

4.3. Municipal controls

In all of our estimations we control for a broad range of variables at the municipal level that can affect health outcomes, and thus omitting them may lead to biased estimates of our coefficients of interest. These controls come from a municipal panel constructed by CEDE (Research Center for Economics and Development at Universidad de los Andes), which provides information on economic, geographical and social characteristics at the municipal level in Colombia for our period of study. Some of the variables that we control for include: population, municipal tax revenues, rurality index, government expenditures on health and education, and average monthly rainfall.¹⁸ This last control is included to take into account possible concerns of cross-contamination among the non-sprayed adjacent areas that could be caused by adverse weather conditions.¹⁹ We also include time-invariant municipal controls, such as size in square kilometers. This control will disappear under the individual fixed-effect estimation if the individual resides in the same location (that is, in our non-migrant sample).

Finally, Table 3 compares sprayed and non-sprayed municipalities for our main outcome variables and other relevant covariates at the municipal level. The proportions of dermatological and respiratory diagnoses as well as miscarriages do not display any statistically significant difference between sprayed and non-sprayed municipalities. With respect to other municipal characteristics, there aren't any differences for sex ratio, population, or public health expenditures (referred to as SGP-health per capita). There are minor differences in the rurality index, fractures and accidents, and there are major differences in average age and area in kilometers. In order to make our sample of municipalities comparable, we restrict the sample to a common support with respect to size (area in square kilometers) by excluding very small municipalities (bottom 5th percentile) and big municipalities (top 5th percentile). Regarding difference in age, we check the consistency of this difference by using data from the 2005 population census. We corroborate that municipalities with aerial spraying have a younger population, this reflects that we do not have a selection problem in our data related to the age of the individuals who attend

health services, but that this is a general characteristic of the population. All characteristics that are different are explicitly controlled for in our empirical specification.

5. Health production function and empirical specification

The health production function was first introduced by Grossman (1972) in the economics literature. Under this framework, health is assumed to be a function of several inputs. The first is the baseline health status of the individual, which is closely related to genetic factors. The second input is leisure, understood as time spent on activities that improve health conditions, such as exercising, cooking and sleeping, among others. The third input is medical care and corresponds to goods and services that individuals can purchase in order to improve their health, for example, quality of doctors, hospitals, medicines, vaccines, and healthy food and habits.

While it is extremely difficult to measure the baseline health status, we deal with this by exploiting the panel structure of our data and using individual fixed effects in our estimations. Fixed effects account for all individual characteristics that do not change over our period of study (genetics, previous health investments, health conditions, and so on). This empirical strategy represents a big advantage over the cross-sectional studies that compare unrelated individuals for whom these conditions differ and cannot be accounted for.

Instead of trying to estimate a structural form of the health production function, we estimate a reduced form of the following health production function:

$$h_{imt} = \Gamma(Y_{it}, l_{it}, \mu_i, s_{mt}) \quad (1)$$

where subscript i refers to the individual, m to the municipality of residence,²⁰ and t to the time period. h is the proportion of respiratory or dermatological diagnosis in a given period of time for an individual i that appears in our panel in municipality m and at time t . Y are goods that can affect health; l corresponds to health inputs that require time; μ is an individual health endowment; and s is an environmental shock that, in our case, corresponds to being exposed to the herbicide used in aerial spraying campaigns in the municipality of residence.

For this setup to be understood as a quasi-experiment or, in other words, for aerial spraying to be exogenous, we need to assume

¹⁸ We lack rainfall information for all of the municipalities so this control restricts our sample, yet we believe it to be an important one. The results hold with and without this control.

¹⁹ Hewitt et al., 2009 show that, even under wind conditions of up to 9 km/h and aircraft speed of up to 333 km/h, spraying will deposit glyphosate close to the targeted area.

²⁰ We do not have specific information about the place of residence of the individual, but we infer that the municipality of residence is the municipality of the facility visited. According to a list of health services in 2011, there are 1083 municipalities with such institutions, which correspond to 98.3% of the territory. Therefore, we believe that this assumption lies close to reality.

Table 4
Effects of aerial spraying on the proportion of dermatological and respiratory diagnosis.

Dependent variable:	Dermatologic diagnosis		Respiratory diagnosis	
	(1)	(2)	(3)	(4)
Panel A: Complete sample				
Average spraying	0.0005*** (0.0002)	0.0005*** (0.0002)	0.002*** (0.001)	0.002*** (0.001)
Percentage of the municipality with coca crops		0.041 (0.041)		−0.080 (0.080)
# of observations	39,766,259	39,766,259	39,766,259	39,766,259
R-squared	0.001	0.001	0.001	0.001
# of individuals	8,319,758	8,319,758	8,319,758	8,319,758
Panel B: Sample of municipalities with positive levels of aerial spraying				
Average spraying	0.0005*** (0.0002)	0.0005*** (0.0002)	0.002*** (0.001)	0.002*** (0.001)
Percentage of the municipality with coca crops		0.089 (0.067)		0.0194 (0.138)
# of observations	7,264,691	7,264,691	7,264,691	7,264,691
R-squared	0.001	0.001	0.002	0.002
# of individuals	1,624,420	1,624,420	1,624,420	1,624,420

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Windows of exposure to aerial spraying prior to the consultation are set to 7 days for dermatological diagnoses and 15 for respiratory diagnoses. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

orthogonality between s , environmental shock, and the error term. In this case, we assume that individuals cannot anticipate with certainty the time and extent of exposure to aerial spraying campaigns. By controlling for characteristics at the municipal level and using municipality or individual fixed effects, there is little concern about the systematic variation between spraying and other characteristics that will bias the estimations of our coefficient of interest. One remaining issue for our identification strategy is the possibility of having self-selection in the sample. This concern would be valid if individuals who go to the doctor are different from those who do not go. So, our analysis is only valid for individuals who attend a health service institution. In addition to this general concern for the sample selection, we may also see selective migration if the most susceptible or concerned individuals regarding exposure to the herbicide used in aerial spraying campaigns decide to migrate. Calculations from our sample show that 7.4% of the individuals in our sample appear in different municipalities throughout our study period. To test for selective migration, we dropped all non-migrants from our sample, then we took the first and last municipality where each individual had a medical consultation and compare the average aerial spraying of the last municipality of residence with respect to the first municipality of residence. We find that the migrants in 51% of the cases moved to a municipality with greater aerial spraying and in 49% of the cases moved to a municipality with lower aerial spraying. These numbers suggest that there is no selective migration related to aerial spraying. If there was still some concern about selective migration, the estimated effect, if any, would be a lower bound. As mentioned previously, farmers might relocate their crops once the spraying campaigns start in their current location and any such shifting of the production reduces the chances of finding any effects. In order to check the robustness of our results we run our regressions for non-migrant individuals and compare them with the full sample results. These two samples give similar results in magnitude for dermatological and respiratory diagnoses; this comparison allows us to eliminate the potential concern of such a bias due to selective migration. The similarities in these results are to be expected, given that the windows of exposure and visits to the health services are 7 and 15 days long, leaving little space for migration over such a short period of time. On the other hand, differences between the estimates of the effect of aerial spraying over miscarriages among the total sample and the non-migrant sample might be plausible due to a longer window of exposure, during which migration might indeed happen.

The reduced-form linear relation of our preferred specification of the health production function is the following model:

$$h_{imt} = \beta_0 + \beta_1 s_{mt} + \beta_2 X_{imt} + \beta_3 Z_{mt} + \gamma_{year} + \gamma_{month} + \gamma_i + \varepsilon_{imt} \quad (2)$$

β_1 is the coefficient associated with our variable of interest, the amount of aerial spraying, s , in municipality m at time t . This variable is measured as the average area (in square kilometers)²¹ sprayed during a relevant window of exposure previous to a medical consultation that takes place at time t . X_{imt} and Z_{mt} are individual and municipal controls that vary over time, respectively. Among the individual controls we include age, age squared, gender, and type of health insurance (none, contributory or subsidized health insurance). Health regime coverage needs to be included to take into account endogenous individuals' consultations. Similarly, we include municipal-level controls, including population, area of municipality (in square km), proportion of rural population, per capita tax revenues, per capita public expenditure on health and education services, and proportion of municipal area cultivated with coca crops.²² Including coca cultivation as a control is important for preventing bias in our coefficient of interest. The high correlation between the extent of aerial spraying and coca cultivation at the municipal level, together with the fact that coca cultivation and cocaine production make indiscriminate use of pesticides, herbicides and chemical precursors, may create a spurious relationship between spraying campaigns and health outcomes if we didn't include both spraying and coca cultivation in our estimations. We also include year γ_{year} , and month γ_{month} fixed effects to control for unobservable factors that change over time, such as seasonal patterns of diseases. γ_i corresponds to an individual fixed effect that controls for all unobservable factors varying at the individual level that are constant over time. The individual fixed-effects model in Eq. (2) is our preferred specification. This specification compares the health outcomes for a given individual when exposed

²¹ We construct a variable for aerial spraying normalized by total area to take into account the size of the municipality. There are significant effects for dermatological, but not for respiratory diagnoses. In any case, we always control for the size of the municipality in our regressions.

²² Total coca cultivation in Colombia is measured every year in December (UNODC and Government of Colombia, 2011). Therefore, we calculate the proportion of the municipal area cultivated with coca as the average of the current and previous years' cultivation, which would correspond to the initial and final measurement for area cultivated for the year of study.

Table 5
Effects of aerial spraying on miscarriages.

Panel A: Complete sample				
Miscarriage diagnosis	Complete sample		Municipalities with positive levels of aerial spraying	
	(1)	(2)	(3)	(4)
Average spraying (9 months)	0.12*** (0.045)	0.118*** (0.040)	0.156*** (0.041)	0.165*** (0.035)
Percentage of the municipality with coca crops		3.79*** (1.407)		6.093*** (2.207)
# of observations	2,947,070	2,947,070	585,132	585,132
R-squared	0.026	0.026	0.034	0.034
# of individuals	687,735	687,735	146,429	146,429
Panel B: Non-migrant population sample				
Miscarriage diagnosis	Non-migrants sample		Non-migrants sample with positive levels of aerial spraying	
	(1)	(2)	(3)	(4)
Average spraying (9 months)	0.154*** (0.055)	0.177*** (0.044)	0.168*** (0.046)	0.192*** (0.037)
Percentage of the municipality with coca crops		8.814*** (2.862)		9.30325*** (2.796)
# of observations	2,815,385	2,815,385	555,628	555,628
R-squared	0.027	0.027	0.034	0.035
# of individuals	656,507	656,507	134,159	134,159

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

to different levels of aerial spraying at different moments in time. As explained before, this identification strategy improves over cross-sectional studies that include municipality fixed effects, which require strong assumptions about shocks being equally perceived by individuals or shocks being homogeneous across all individuals in a given municipality. ε_{imt} is an error term assumed to be orthogonal to aerial spraying.

We restrict the time window prior to medical consultation to a given number of days in our estimations; the time window chosen is consistent with previous findings in the medical literature. For dermatological and respiratory diagnoses we use a time window of 7 and 15 days before medical consultation, respectively. For miscarriages, we use eight months prior to the last prenatal care visit (to account for three months prior to conception and the first five months of pregnancy when a loss is considered a miscarriage) and an additional month to account for some time when the miscarriage might occur after the last prenatal care visit. This gives a total time window of nine months. In order to check the robustness of our results with respect to dermatological problems, we estimate the individual fixed-effects model in Eq. (2) for time windows of 3–15 days and the results are consistent throughout the different specifications. Once we use greater time windows of 30, 45 and 60 days the effects of aerial spraying on dermatological and respiratory conditions fade, which reinforces the fact that we identify a causal effect consistent with the timing reported by the medical literature.

6. Results

We use three different dependent variables that correspond to medical diagnoses identified in the medical literature as being related to exposure to glyphosate in the short run. Table 4 presents our baseline estimations of Eq. (2) for dermatological and respiratory diagnoses. Columns 1 and 2 correspond to dermatological diagnosis and columns 3 and 4 to respiratory diagnosis. Results in all columns include individual fixed effects, year fixed effects and

all individual and municipal controls.²³ Columns 1 and 3 do not include as a municipal control the amount of coca cultivation in the municipality of residence, whereas columns 2 and 4 do.

Table 4 presents the main results estimated from Eq. (2) in Panel A for the full sample (including all municipalities in Colombia) and in Panel B for the sample of municipalities that had positive levels of spraying at any moment in the period of study. The second panel accounts for concerns related to the differences among sprayed and non-sprayed areas by looking at the effects of aerial spraying on health outcomes at the intensive margin.

Regarding the dependent variable, we calculate the proportion of dermatological and respiratory diagnoses in the last 7 and 15 days respectively, with respect to the total number of visits that the individual made during the current year. Including a measure of diagnoses as a proportion of total visits to the doctor eliminates any bias related to the fact that not everyone goes to the doctor with the same frequency.

The first column in Panel A of Table 4 shows that, on average, a one square kilometer increase in the area sprayed with glyphosate increases the proportion of dermatological diagnosis in the 7-day window following exposure to the herbicide by 0.00045. This effect is statistically significant at the 1% confidence level. When we include the extent of coca cultivation (column 2) as a control, our coefficient of interest remains the same and is still significant at the 1% level. If we take the result of our preferred specification in column 2, this implies that a one standard deviation increase in the area sprayed with glyphosate in a 7-day time window prior to medical consultation increases the proportion of dermatological consultations by 0.35% (from a baseline average of 1.2%). Although this effect seems small, it is highly significant.

Columns 3 and 4 in Panel A of Table 4 present our baseline results for the case of respiratory diagnosis. The results indicate that a one square kilometer increase in the area sprayed with glyphosate

²³ Once we include the individual fixed effects, the results are not very sensitive to the sequential inclusion of other control variables. Tables available upon request.

Table 6

Effects of aerial spraying on the proportion of dermatological and respiratory diagnosis in high-income and low-income municipalities.

Dependent variable:	Dermatologic diagnosis		Respiratory diagnosis	
	(1)	(2)	(3)	(4)
Panel A: High income municipalities ^a				
Average spraying	0.005*** (0.001)	0.005*** (0.001)	0.007* (0.004)	0.007* (0.004)
Percentage of the municipality with coca crops		0.578 (1.569)		−0.694 (4.785)
# of observations	20,131,375	20,131,375	20,131,375	20,131,375
R-squared	0.001	0.001	0.001	0.001
# of individuals	4,326,129	4,326,129	4,326,129	4,326,129
Panel B: Low income municipalities ^b				
Average spraying	0.0004*** (0.0001)	0.0004*** (0.0001)	0.002** (0.001)	0.002** (0.001)
Percentage of the municipality with coca crops		0.023 (0.048)		−0.085 (0.100)
# of observations	19,634,884	19,634,884	19,634,884	19,634,884
R-squared	0.001	0.001	0.001	0.001
# of individuals	4,354,915	4,354,915	4,354,915	4,354,915

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Windows of exposure to aerial spraying prior to the consultation are set to 7 days for dermatological diagnoses and 15 for respiratory diagnoses. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

^a We define high-income municipalities as those above the 50th percentile of tax income collected by the municipality.

^b We define low-income municipalities as those below the 50th percentile of tax income collected by the municipality.

increases the proportion of respiratory diagnosis in the 15-day window following exposure to the herbicide by 0.00218. The effect on the proportion of respiratory diagnosis is statistically significant at the 1% confidence level. This result is robust and does not change significantly once we include the extent of coca cultivation in the municipality of residence as a control (column 4). If we take the result from column 4, a one standard deviation increase in aerial spraying leads to an increase in the proportion of medical consultations with respiratory diagnosis by 0.465% (from a baseline average of 3.6%).

Panel B in Table 4 presents the results when we restrict our sample to those municipalities that had positive levels of aerial spraying at any moment during our period of study. Therefore, we exploit the intensive margin of the quasi-experiment. This panel shows that the results obtained when using only those municipalities with positive levels of aerial spraying go in the same direction as the results obtained when using the whole sample. The estimated coefficient implies that a one standard deviation increase in the area sprayed with glyphosate increases the proportion of dermatological consultations by 0.85% and by 0.87% in the case of respiratory diagnosis. In other words, when we restrict our sample to municipalities that had positive levels of aerial spraying, the effect of spraying on the proportion of medical consultations related to dermatological problems remains small, but is 2.4 times larger than that obtained using all of the municipalities in Colombia. This proportion is 1.87 in the case of respiratory diagnosis.

Table 5 presents the results for miscarriages. As mentioned previously, we use eight months prior to the last prenatal care visit and one month later (due to a lack of certainty regarding the exact date of the miscarriage), which gives a total time window of nine months. In the case of miscarriages, we focus on individuals who did not change municipality of residence, since not doing so might generate measurement error in the extent of exposure to the herbicide used in the aerial spraying campaigns, resulting in estimates that would be a lower bound of the effect. In addition, not knowing the exact date of migration (we only observe the municipality of residence once an individual goes to a health service institution) prevents us from being able to assign the true level of exposure to glyphosate that a migrant woman faced during her pregnancy and

in her different municipalities of residence.²⁴ This bias comes from the fact that there could be selective migration, as individuals who were exposed to aerial spraying would move to areas with lower levels of aerial spraying, and the number of diagnoses related to glyphosate will average out for the most part.

The results in Table 5 show that aerial spraying has a strong and statistically significant effect on miscarriages, and indeed, a larger effect for the non-migrant sample. As mentioned before, our measure of miscarriages might be overestimated. For this reason, we only analyze the estimated coefficient as proportions of the baseline rate. The results from Table 5 imply that a one standard deviation increase in aerial spraying increases the rate of miscarriages by 2.9% across the entire sample of municipalities, and by 8.7% in municipalities with positive levels of aerial spraying in the full sample. The results obtained for miscarriages when we use the non-migrant sample of individuals (Panel B) imply that a one standard deviation increase in aerial spraying increases miscarriages by 4.4% across the entire sample of municipalities, and by 10.4% in municipalities with positive levels of aerial spraying. All estimated effects for the case of miscarriages are statistically significant at the 1% confidence level.

6.1. Robustness checks

6.1.1. High and low income municipalities

One concern that may arise from the results presented in Tables 4 and 5 is their external validity due to the quality of the data. One way to partially address this is to estimate the same group of regressions by dividing the sample between individuals living in low-income and high-income municipalities. If results are consistent and homogeneous in both samples, these alleviate the concern of lack of external validity or differential quality of the data reported by low and high income municipalities. We define high-income municipalities as those above the 50th percentile of tax income collected by the municipality. Table 6 presents the results of our estimations for the case of the proportion of dermatologi-

²⁴ Although we show robustness checks using the non-migrant sample for the case of dermatological and respiratory diagnoses, this is less of a concern for these two conditions given that we use 7-day and 15-day time windows between exposure to spraying campaigns and medical consultation.

Table 7
Effects of aerial spraying on miscarriages in high-income and low-income municipalities.

Miscarriage diagnosis	Non-migrant sample, low income municipalities		Non-migrant sample, high income municipalities ^a	
	(1)	(2)	(3)	(4)
Average spraying (9 months)	0.146*** (0.049)	0.171*** (0.039)	3.33* (1.710)	3.372* (1.776)
Percentage of the municipality with coca crops		9.407*** (2.831)		–35.937 (72.394)
# of observations	1,400,193	1,400,193	1,415,192	1,415,192
R-squared	0.030	0.031	0.030	0.030
# of individuals	347,512	347,512	320,083	320,083

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

^a We define high-income municipalities as those above the 50th percentile of tax income collected by the municipality and low-income municipalities as those below the 50th percentile of tax income collected by the municipality.

Table 8
Effects of aerial spraying on the proportion of dermatological and respiratory diagnosis (Non-migrant population sample).

Dependent variable:	Dermatologic diagnosis		Respiratory diagnosis	
	(1)	(2)	(3)	(4)
Average spraying	0.0005*** (0.0002)	0.0005*** (0.0002)	0.002*** (0.0008)	0.002*** (0.0008)
Percentage of the municipality with coca crops		0.0006 (0.10)		0.04 (0.17208)
# of observations	37,736,485	37,736,485	37,736,485	37,736,485
R-squared	0.001	0.001	0.001	0.001
# of individuals	7,972,585	7,972,585	7,972,585	7,972,585

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Windows of exposure to aerial spraying prior to the consultation are set to 7 days for dermatological diagnoses and 15 for respiratory diagnoses. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

cal and respiratory diagnosis when we divide the sample between high-income municipalities (Panel A) and low-income municipalities (Panel B). The two panels in this table have the same structure as the results presented in Table 4, that is, columns 1 and 3 present the results without controlling for the level of coca cultivation in each municipality for dermatological and respiratory diagnosis respectively, while columns 2 and 4 control for the extent of coca cultivation.

The estimated effects of aerial spraying on dermatological diagnoses are significant at the 1% confidence level for both high-income and low-income municipalities, and the effects on respiratory diagnoses are significant at the 10% and 5% confidence levels in high-income and low-income municipalities, respectively. In the case of high-income municipalities, a one square kilometer increase in the area sprayed with glyphosate increases the proportion of dermatological diagnoses in the 7-day window following exposure to the aerial spraying campaigns by 0.0047 in our preferred specification, and by 0.0067 in the case of respiratory diagnosis. In low-income municipalities, the estimated effects on dermatological and respiratory diagnoses are 0.00038 and 0.00206, respectively. In order to compare the economic significance of the results we should take into account the baseline levels of aerial spraying in the two groups of municipalities. For the group of high-income municipalities, the results imply that a one standard deviation increase in the area sprayed with glyphosate results in a 0.5% increase in the proportion of dermatological diagnosis in the following 7 days after the spraying event (from a baseline average of 1.2%) and a 0.17% increase in the proportion of respiratory diagnosis (from a baseline average of 3.4%). The results for low-income municipalities reported in Panel B imply that a one standard deviation increase in the area sprayed with glyphosate results in a 0.42% increase in the proportion of dermatological diagnosis in the following 7 days after the spraying event (from a baseline average of

1.2%), and a 0.58% increase in proportion of respiratory diagnosis (from a baseline average of 3.9%).

Table 7 presents the results for the case of miscarriages as the dependent variable when we divide the sample between low-income and high-income municipalities, again using the sample of non-migrant individuals. In this case, the effects of aerial spraying on miscarriages are statistically significant at the 1% and 10% confidence levels for low-income and high-income municipalities, respectively. Columns 1 and 2 present the results for low-income municipalities and columns 3 and 4 for high-income municipalities. These results show that a one standard deviation increase in aerial spraying increases miscarriages by 5.8% in low-income municipalities and by 3.7% in high-income municipalities.

6.1.2. Non-migrants for dermatological and respiratory diagnoses

In order to show that our results for the proportion of dermatological and respiratory diagnosis are not biased by selective migration (if it happens to be the case that the individuals most susceptible to the effects of exposure to the herbicide decide to migrate in order to avoid the consequences of exposure to the herbicide), Table 8 presents the results for estimating Eq. (2) using only the non-migrants subsample of individuals. The results in Table 8 confirm all previous findings. Namely, that aerial spraying has a positive and significant effect on the proportion of dermatological and respiratory diagnoses. According to the results from columns 2 and 4 shown in Table 8, a one standard deviation increase in aerial spraying increases the proportion of dermatological diagnosis by 0.38% (from a baseline average of 1.2%), and the proportion of respiratory diagnosis by 0.48%. These results are significant at the 1% confidence level.

6.1.3. Different windows of exposure and non-linearities

We also test whether the effects of aerial spraying campaigns are robust to variations in the window of exposure found in the

Table 9

Effects of aerial spraying on the proportion of dermatological and respiratory diagnosis for 30 and 45 days' time window.

Dependent variable:	Dermatologic diagnosis	Respiratory diagnosis
	(1)	(2)
Panel A: 30 days time window		
Average spraying (30 days)	0.0002 (0.0002)	0.003** (0.001)
Percentage of the municipality with coca crops	0.042 (0.041)	−0.08 (0.079)
# of observations	39,766,259	39,766,259
R-squared	0.001	0.001
# of individuals	8,319,758	8,319,758
Panel B: 45 days time window		
Average spraying (45 days)	0.0005* (0.0002)	0.002 (0.001)
Percentage of the municipality with coca crops	0.042 (0.041)	−0.078 (0.079)
# of observations	39,766,259	39,766,259
R-squared	0.001	0.001
# of individuals	8,319,758	8,319,758

Standard errors clustered at the municipality level in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

Table 10

Placebo test – Effects of aerial spraying on the proportion of accidents and fractures.

Panel A: Simple sample				
Dependent variable:	Accidents diagnosis		Fractures diagnosis	
	(1)	(2)	(3)	(4)
Average spraying (15 days)	0.0003 (0.0002)	0.0003 (0.0002)	0.0002 (0.0002)	0.0002 (0.0002)
Percentage of the municipality with coca crops		−0.068*** (0.026)		−0.056** (0.025)
# of observations	39,766,259	39,766,259	39,766,259	39,766,259
R-squared	0.004	0.004	0.004	0.004
# of individuals	8,319,758	8,319,758	8,319,758	8,319,758

Standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include the following controls: age, age square, health regime, municipal tax income, population, area in square km, rurality index, average monthly rainfall, municipal spending on education and health, subsidized regime coverage, year and month dummy.

literature. For dermatological events, symptoms might appear 24 h after exposure. The main results presented in this paper use a longer time window of 7 days for dermatological events to account for the time that a person takes to acknowledge that he/she has a medical issue and plans to go to the doctor, but the results are positive and significant for any time window between 3 and 15 days.

We also perform an empirical analysis for dermatological and respiratory diagnoses using windows of exposure that last beyond the previously explored 15-day time window. Table 9 presents the results for estimating Eq. (2) using time windows of 30 days (Panel A) and 45 days (Panel B) between exposure to the spraying campaigns and the associated medical consultation. This table presents the results of our preferred specifications, where we control for the extent of coca cultivation in each municipality. In the case of dermatological diagnosis, the effects of exposure to the spraying campaigns within 30 days prior to medical consultation are not statistically significant, whereas for respiratory diagnosis the effect is positive and statistically significant at the 5% confidence level. In Panel B, where we explore the effects over a 45-day time window, the effects are again positive but only statistically significant at the 10% level for dermatological diagnosis. As expected after consulting the literature, the results presented in Table 4 and compared to Table 9 show that the effects of exposure to glyphosate are highly significant within the relevant window of exposure, yet these results lose their statistical precision once we adopt a longer period of exposure.

Strong evidence remains absent for the existence of non-linearity among the impact of aerial spraying on dermatological

and respiratory diagnoses or miscarriages. The linear specification in our variable of interest should be the strongest one, as we are estimating an individual fixed effect model. On average we observe each individual 4.7 times in the five-year period of study. Therefore, the within variance in the size of exposure to glyphosate among the same individual is not high enough to identify a non-linear effect.²⁵

6.1.4. Placebo test: accidents and fractures

Finally, we present the results of a placebo test where we use as the dependent variable the proportion of two conditions that, in principle, should be completely unrelated to exposure to the spraying campaigns: accidents and fractures.²⁶ The results presented in Table 10 show that exposure to the spraying campaigns within a 15-day time window does not lead to an increase in the fraction of medical consultations related to these two conditions. These results show that the estimated effects obtained in Tables 4–9 do not come from the high statistical power obtained, but from a causal relationship between exposure to glyphosate used in the aerial spraying program in Colombia and some medical conditions.

²⁵ We do not include the results in this paper, but they are available upon request.

²⁶ We define accidents by using the following categories from the ICD10 diagnoses: traumas and fractures (codes starting with S); injuries and collisions (V codes); falls, bites or stings (W codes); burns or poisoning (X codes). Bone fractures represent some specific codes with the S category, so comprise a subset of accidents.

7. Concluding remarks

This paper evaluates the effects that the aerial spraying of herbicides to reduce illicit crop cultivation has on health outcomes in Colombia. Using a unique dataset of individual-level health registries, together with daily information regarding the extent of aerial spraying campaigns at the municipal level, we estimate individual fixed-effects regressions that represent the closest counterpart to an experiment. Our results indicate that exposure to the herbicides used in aerial spraying campaigns leads to an increase in dermatological and respiratory problems and miscarriages. The estimated effects of aerial spraying on these three medical conditions are highly significant and robust to different subsamples and specifications.

The results in our paper imply that the aerial spraying of herbicides as a strategy to reduce coca cultivation in Colombia has a negative collateral effect on the health outcomes of populations exposed to the herbicide. Other papers in the economics literature show that aerial spraying campaigns are ineffective at reducing coca cultivation. The most conservative (and positive) estimates of the effects of aerial spraying on coca cultivation indicate that for each additional hectare sprayed with herbicides, coca cultivation is reduced by about 0.035 ha (Mejía et al., 2015). This implies that in order to reduce the amount of coca cultivation by one hectare, almost 30 ha would have to be sprayed. Furthermore, the same authors estimate that the average cost of eliminating one hectare of coca crops through the aerial spraying program is about USD\$74,000.

While this version of the paper was being completed, the Colombian government announced that it would stop the aerial spraying program. The decision was taken based on the possible health effects that the program might be having on populations exposed to the herbicide used in the aerial spraying campaigns. The findings in this paper indicate that it is indeed the case that the aerial spraying program has negative health consequences.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jhealeco.2017.04.005>.

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